

Exp No: 5

Arduino based Manual Electronic Counter using Proteus

OBJECTIVES

- To design a manual electronic counter using Arduino UNO.
- To interface a push button and 7-segment display with Arduino.
- To simulate and verify the 0–9 counting operation using Proteus.

PROGRAM

```
const int buttonPin = 12;

const int seg[7] = {7, 6, 5, 4, 3, 2, 1};

// 7-segment patterns for 0 to 9
byte numbers[10][7] = {

  {1,1,1,1,1,1,0}, // 0
  {0,1,1,0,0,0,0}, // 1
  {1,1,0,1,1,0,1}, // 2
  {1,1,1,1,0,0,1}, // 3
  {0,1,1,0,0,1,1}, // 4
  {1,0,1,1,0,1,1}, // 5
  {1,0,1,1,1,1,1}, // 6
  {1,1,1,0,0,0,0}, // 7
  {1,1,1,1,1,1,1}, // 8
  {1,1,1,1,0,1,1} // 9
};

int count = 0;

int lastButtonState = HIGH;

void setup()

{

  for (int i = 0; i < 7; i++)
```

```

{
    pinMode(seg[i], OUTPUT);
}

pinMode(buttonPin, INPUT_PULLUP);

displayNumber(0);
}
void loop()
{
    int buttonState = digitalRead(buttonPin);

    if (lastButtonState == HIGH && buttonState == LOW)
    {
        count++;

        if (count > 9)
        {
            count = 0;
        }

        displayNumber(count);
        delay(200);
    }

    lastButtonState = buttonState;
}
void displayNumber(int number)
{
    for (int i = 0; i < 7; i++)

```

```

{
    digitalWrite(seg[i], numbers[number][i]);
}
}

```

OUTPUT

